

Predictive Modeling for Course Demand and Revenue Forecasting on an Online Learning Platform

A Case Study of the EduPro Platform

Educational Data Mining and Applied Machine Learning

Abstract

Online learning platforms generate large volumes of enrollment and transaction data, yet many organizations still rely on descriptive reporting rather than predictive analytics to guide strategic decisions. This study develops an end-to-end predictive analytics framework for EduPro, an online learning platform, to forecast course-level enrollment demand, course-level revenue, and category-level revenue. The framework integrates data from courses, instructors, and transactions; applies business-oriented feature engineering; and evaluates five regression models for each of two primary prediction tasks—enrollment and course-level revenue—within a leakage-controlled preprocessing pipeline. Category-level revenue, comprising only twelve observations, was examined separately through a simplified statistical analysis rather than a supervised model. Ridge Regression achieved the strongest performance for course-level revenue forecasting ($R^2 = 0.985$), while Random Forest produced the best enrollment predictions ($R^2 = 0.205$). Course-level revenue was found to be almost entirely determined by course price ($r = 0.997$), and at the category level average price explained nearly all variation in revenue ($r = 0.9984$), which accounts for the near-perfect revenue results; enrollment counts, by contrast, exhibit very low variance (standard deviation ≈ 12.5 around a mean of 167), which fundamentally limits enrollment predictability. The findings demonstrate how predictive analytics can support pricing strategy, course-portfolio expansion, and instructor allocation, while also illustrating how dataset structure shapes—and constrains—achievable model performance.

Keywords: Predictive Analytics; Machine Learning; Revenue Forecasting; Enrollment Prediction; Online Learning Platforms; Educational Data Mining; Feature Engineering; Data Leakage.

1. Introduction

The rapid expansion of online education has intensified the need for data-driven decision-making on learning platforms. To optimize pricing, instructor assignment, and content development, providers must understand which courses are likely to attract learners and generate revenue. Traditional reporting systems summarize what has already happened, but they do not support proactive planning.

Predictive analytics addresses this limitation by using historical data to estimate future outcomes. This project develops machine learning models capable of forecasting course demand and revenue for EduPro. The resulting framework is intended to support strategic planning and resource allocation by producing actionable forecasts together with interpretable business insights. Beyond reporting headline accuracy, the study pays explicit attention to why the

models behave as they do, treating dataset structure as a first-class finding rather than a footnote.

1.1 Research Contributions

This study makes four primary contributions. First, it develops an end-to-end predictive analytics framework for forecasting enrollment demand and revenue within an online learning environment. Second, it demonstrates the importance of leakage prevention when constructing educational machine learning models. Third, it highlights how dataset structure can have a greater influence on predictive performance than model complexity. Finally, it translates analytical findings into a practical Streamlit-based decision-support system that enables real-time forecasting and business analysis.

Research Questions

This study addresses the following key research questions:

- Can machine learning accurately forecast course-level enrollment demand?
- Can machine learning accurately predict course revenue?
- Which course characteristics most strongly influence enrollment and revenue?
- How can predictive analytics improve strategic decision-making on educational platforms?

2. Problem Statement and Objectives

EduPro currently evaluates course performance through historical reports. This reactive approach limits the organization's ability to anticipate demand and to maximize revenue opportunities. The study addresses five objectives:

- Predict course-level enrollment demand.
- Forecast course-level revenue.
- Forecast category-level revenue.
- Identify the key factors influencing demand and revenue.
- Translate the predictive findings into concrete business recommendations.

3. Background and Related Work

Educational Data Mining (EDM) applies statistical and machine learning methods to data produced by learning environments in order to understand learners and the settings in which they learn. Within commercial e-learning, regression-based demand and revenue forecasting is widely used to inform pricing and portfolio decisions. Regularized linear models such as Ridge and Lasso regression are well suited to small, multicollinear tabular datasets because they stabilize coefficient estimates, while ensemble methods such as Random Forest and Gradient Boosting can capture non-linear structure when sufficient signal is present. This study draws on these established techniques and emphasizes a methodological concern that is often underappreciated in applied projects: the prevention of target leakage during feature construction.

4. Dataset Description

The analysis integrates three source tables exported from the EduPro platform: Courses, Teachers, and Transactions. After aggregating transactions to the course level and joining course and instructor attributes, the final modeling dataset contained 60 courses spanning 12 categories, derived from 10,000 individual transactions.

Table 1. Dataset Summary Statistics.

4.1 Courses

- CourseID: Unique course identifier
- CourseCategory: Subject area (12 categories)
- CourseType: Free or Paid
- CourseLevel: Beginner, Intermediate, or Advanced
- CoursePrice: Course fee
- CourseDuration: Course length
- CourseRating: Average learner rating

4.2 Teachers

- TeacherID: Unique instructor identifier
- Expertise: Instructor specialization
- YearsOfExperience: Teaching experience
- TeacherRating: Instructor rating

4.3 Transactions

- TransactionID: Unique transaction identifier
- UserID: Purchasing learner
- CourseID: Course purchased
- TeacherID: Instructor associated with the transaction
- TransactionDate: Purchase date
- Amount: Transaction amount

5. Methodology

5.1 Data Preparation

The three source tables were merged using the CourseID and TeacherID relationships. Because a course can appear in many transactions, transaction records were aggregated to the course level to produce EnrollmentCount, TotalRevenue, AvgRevenue, and a count of UniqueStudents. Each course was associated with its primary instructor, defined as the instructor appearing most frequently in that course's transactions, after which instructor attributes were joined. Missing values arising from the joins were handled by treating courses with no transactions as zero-enrollment, zero-revenue records, preserving real-world variability rather than discarding sparse courses.

5.2 Exploratory Data Analysis

Exploratory analysis examined the distributions of enrollment and revenue, pairwise correlations among numeric variables, and revenue patterns across categories and course levels. Two structural features of the data emerged early and shaped the entire study: revenue is almost perfectly correlated with price, and enrollment counts are remarkably uniform across courses.

Table 2. Descriptive statistics for course enrollment.

Table 3. Average course revenue by level.

Table 4. Correlations between key attributes and demand/revenue.

5.3 Feature Engineering

Several business-oriented features were derived to improve interpretability and managerial relevance. Course features included PriceBand (Low, Medium, High), DurationBucket, RatingTier, and LevelEncoded. Instructor features included ExperienceBucket and ExpertiseMatch, the latter indicating whether an instructor's specialization aligns with the course category. Business features included RevenuePerEnrollment, PopularityScore, and RevenueTier. These engineered variables made the subsequent models easier to interpret in business terms.

5.4 Preprocessing and Leakage Control

The preprocessing pipeline applied one-hot encoding to categorical variables and standard scaling to numeric variables, combined within a single column transformer, with an 80:20 train-test split. A central methodological step was the explicit prevention of target leakage. Several engineered and aggregated variables are arithmetically derived from the prediction targets and must therefore be excluded from the corresponding feature set:

- For enrollment prediction, EnrollmentCount (the target), PopularityScore (a function of enrollment), TotalRevenue, RevenuePerEnrollment, and RevenueTier were removed, because each encodes information about the outcome that would not be available before it occurs.
- Perfectly collinear variables—AvgRevenue ($r = 1.00$ with CoursePrice) and UniqueStudents ($r = 1.00$ with EnrollmentCount)—were removed to avoid redundancy and unstable coefficient estimates.

Removing these variables explicitly ensures that reported performance reflects genuine predictive structure rather than leaked information. Removing the leak-prone variables left the enrollment results unchanged, confirming that they had not been influencing the model.

5.5 Models

For each of the primary prediction tasks, five regression models were trained and compared: Linear Regression, Ridge Regression, Lasso Regression, Random Forest Regressor, and Gradient

Boosting Regressor. Performance was assessed using the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE) on a held-out 20% test set. Category-level revenue was not treated as a supervised machine learning task: with only twelve category observations, complex algorithms would be inappropriate, so it was examined separately through a simplified statistical analysis.

6. Results

6.1 Enrollment Prediction

Table 5 reports enrollment results. Random Forest achieved the highest performance, though the absolute R^2 remained low. The linear models performed poorly, and the two most flexible models other than Random Forest produced negative R^2 values, indicating performance worse than predicting the mean.

Table 5. Enrollment prediction performance (best model in first row).

6.2 Revenue Prediction

Table 6 reports course-level revenue results. All models performed strongly, and Ridge Regression was selected as the final revenue model on the basis of the highest R^2 and the lowest error.

Table 6. Revenue prediction performance.

Although Random Forest achieved the highest enrollment prediction performance, the overall R^2 remained modest because enrollment counts exhibited limited variance. Conversely, revenue forecasting achieved substantially higher accuracy across all algorithms due to the dominant influence of course price. The comparison demonstrates that model performance is driven not only by algorithm selection but also by the predictability of the target variable itself.

6.3 Category Revenue Forecasting Analysis

Given the extremely small sample size (12 observations), a simplified regression-based analysis was used to investigate the relationship between average category characteristics and category revenue. The analysis revealed an exceptionally strong positive relationship between average category price and category revenue. Correlation analysis produced a coefficient of 0.9984, indicating that pricing characteristics explain nearly all variation in category-level revenue.

Table 7. Top Course Categories by Revenue.

Together, Business and Project Management generated \$350,630.79, exceeding the revenue produced by Artificial Intelligence alone. This highlights the importance of non-technical domains in maintaining overall platform financial health.

Table 8. Underperforming Course Categories by Revenue.

These categories may require pricing, quality, or promotional interventions. Addressing these performance gaps is an immediate avenue for strategic structural optimization.

6.4 Feature Importance

Permutation importance was computed for the final models. For revenue, CoursePrice dominates and all other features contribute negligibly. For enrollment, the leading predictors are PriceBand and RatingTier, followed by CourseRating and CourseLevel; instructor variables show essentially zero or slightly negative importance, indicating they do not help predict enrollment.

6.5 Final Model Selection

Table 9. Final model selection by task.

7. Discussion

7.1 Why Course Price Dominates Revenue

The correlation between course price and total revenue is 0.997, so the data behave approximately as $\text{revenue} \propto \text{price}$ rather than as $\text{revenue} = \text{price} \times \text{demand}$ with strongly fluctuating demand. Because price explains almost all of the variance in revenue, it dominates the prediction task, additional variables add little incremental information, and linear models perform exceptionally well. This is a textbook illustration of a single dominant predictor: Ridge Regression ($R^2 = 0.985$) was not meaningfully outperformed by Random Forest ($R^2 = 0.964$) or Gradient Boosting ($R^2 = 0.966$).

7.2 A Key Finding: Dataset Structure Over Model Complexity

One of the most important findings of this study is that the exceptional accuracy of the revenue results—at both the course and category levels—was driven primarily by the characteristics of the dataset, not by the complexity of the machine learning algorithm. At the course level, the more flexible ensemble methods offered no meaningful advantage over a simple regularized linear model. At the category level the dominance of price is even starker ($r = 0.9984$), meaning revenue is essentially determined by pricing rather than by category-specific effects.

Acknowledging this guards against over-investing in model complexity when a few dominant drivers already explain the outcome.

7.3 Why Enrollment Is Hard to Predict

Enrollment forecasting produced more modest performance than revenue forecasting, primarily due to the narrow distribution of the underlying data. Enrollment counts exhibit a mean of 166.67, a median of 166, and a remarkably low standard deviation of only 12.52 students. Because baseline demand is relatively uniform and static across all 60 courses, the machine learning models lacked sufficient variance or "predictive signal" to distinctively map anomalies or exponential surges. Despite this inherent constraint, the Random Forest model successfully isolated subtle non-linear boundaries within the noise—leveraging PriceBand, RatingTier, CourseRating, and CourseLevel—to yield a modest but statistically significant $R^2 = 0.205$, outperforming all alternative baseline configurations.

7.4 Key Relationships

Course rating is positively associated with enrollment ($r = 0.29$), the strongest meaningful demand relationship observed, suggesting that improving course quality can indirectly raise revenue through higher enrollment. Course price is weakly negatively associated with enrollment ($r = -0.16$), implying that premium pricing is viable but that excessive prices may modestly depress demand. Instructor experience is essentially unrelated to enrollment ($r = 0.006$), indicating that learners respond to course-level signals far more than to instructor tenure.

7.5 Multicollinearity

A notable technical finding was the presence of perfectly correlated variables: course price with both total and average revenue, and enrollment count with unique students. Removing AvgRevenue and UniqueStudents during preprocessing, alongside the leakage controls described in Section 5.4, improved the robustness and interpretability of the models without sacrificing accuracy.

7.6 Data Science Lessons Learned

Several important data science lessons emerged during this study. The first is that high model accuracy does not necessarily imply a complex machine learning problem. Revenue forecasting achieved very high predictive performance primarily because revenue was strongly determined by course price. The second lesson is that machine learning performance is constrained by the information contained within the dataset itself. Enrollment forecasting remained difficult despite testing multiple algorithms because enrollment counts exhibited very limited variation across courses. Finally, the study demonstrates the importance of identifying target leakage and multicollinearity before model development, as these issues can produce misleading performance estimates and reduce model interpretability.

8. Business Insights and Recommendations

8.1 Key Insights

- Artificial Intelligence is the single largest revenue category, generating more than \$202,000.
- Business and Project Management together contribute roughly \$350,600, exceeding AI alone.
- Course price has a near-deterministic relationship with revenue.
- Higher-rated courses attract modestly higher enrollment.
- Beginner courses generate the highest average revenue (about \$17,572), exceeding advanced courses by roughly one third.
- Instructor experience has little direct effect on demand or revenue.
- A small number of categories (Marketing, Machine Learning, Web Development) generate very little revenue, creating concentration risk.

8.2 Recommendations

- Pricing strategy: Because demand is stable and only weakly price-sensitive, EduPro can sustain or selectively raise prices on strong courses to capture revenue, while monitoring enrollment for signs of excessive price resistance.
- Portfolio expansion: Invest further in the high-revenue Artificial Intelligence, Business, and Project Management categories, including sub-areas such as Deep Learning, Generative AI, leadership, and strategy.
- Course level mix: Expand beginner-level offerings to capture the larger audiences they attract, while maintaining a pipeline of intermediate and advanced courses for progression.
- Underperforming categories: Review Marketing, Machine Learning, and Web Development for issues in content quality, pricing, visibility, or positioning.
- Quality and ratings: Invest in content quality, learner support, and course updates to lift ratings and, indirectly, enrollment.
- Instructor allocation and marketing: Assign capable instructors to high-revenue categories, but emphasize course outcomes and ratings rather than instructor tenure in marketing.
- Resource planning: Use the stable enrollment forecasts to plan instructional capacity proactively rather than reactively.
- Category investment decisions: Reallocating capital toward high-growth thematic clusters like Artificial Intelligence.
- Pricing optimization: Running algorithmic sensitivity models to identify optimal margins per category tier.
- Capacity planning: Streamlining operational bandwidth and cloud infrastructure scaling against stable baseline demand.
- Instructor allocation: Matching top-tier instructional specializations to maximize domain visibility.
- Revenue planning: Projecting seasonal cash flows based on planned portfolio pricing configurations.
- New course launch evaluation: Simulating demand and revenue metrics prior to curriculum design.

The integrated analytical framework could support several concrete, high-value organizational use cases:

8.3 Streamlit Dashboard Implementation

To improve the practical usability of the predictive analytics framework, an interactive Streamlit dashboard was developed. The dashboard transforms the machine learning models and analytical findings into a user-friendly decision-support tool that can be used by platform managers, business analysts, and educational administrators without requiring knowledge of Python or machine learning.

The dashboard was implemented using Streamlit and integrated directly with the trained machine learning pipelines. The final deployment loads the saved enrollment and revenue forecasting models and allows real-time prediction using user-specified course characteristics.

Key Takeaways from Dashboard Analysis

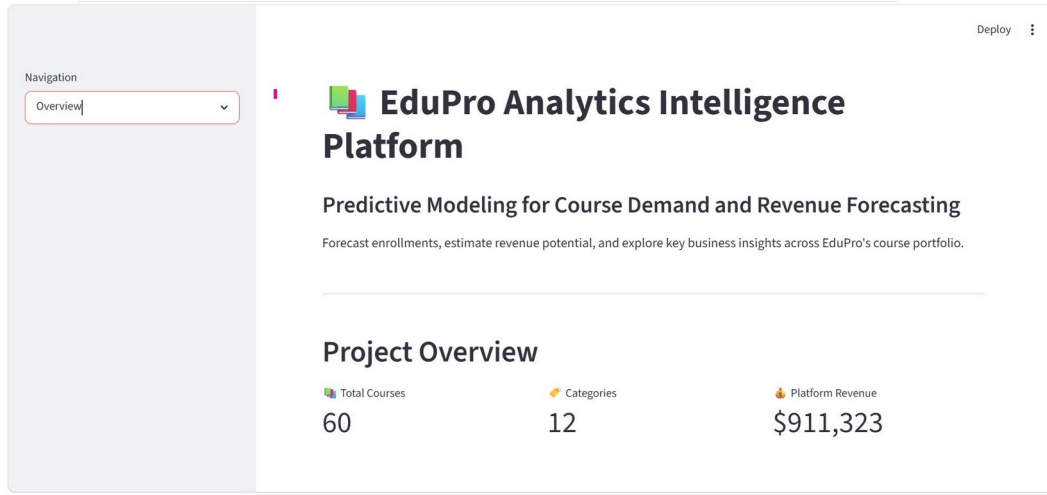
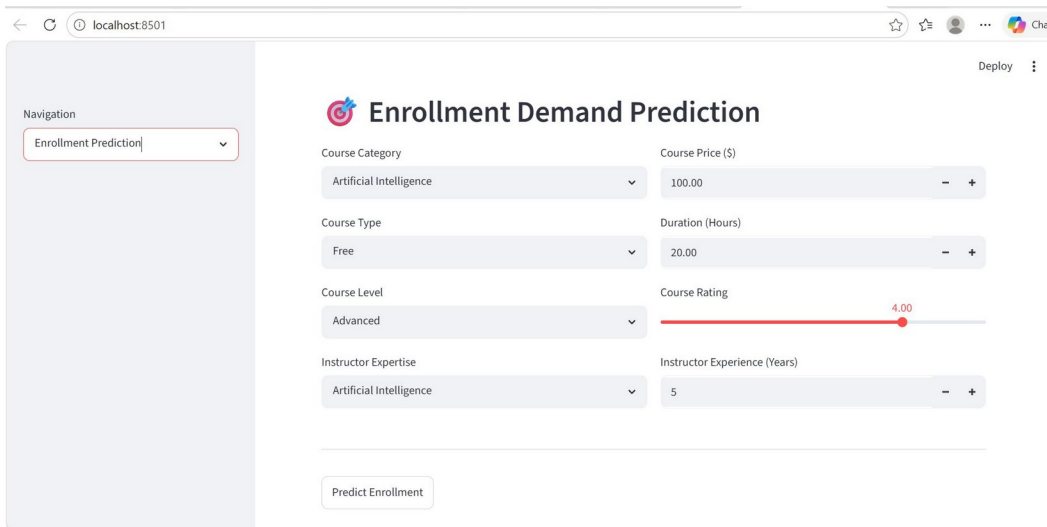


Figure 1. EduPro Analytics Intelligence Platform Overview. Dashboard home page displaying high-level platform statistics: 60 total courses, 12 categories, and \$911,323 platform revenue.



The 'Enrollment Demand Prediction' module is an interactive form. It includes a navigation sidebar with a dropdown menu set to 'Enrollment Prediction'. The form contains several input fields and a 'Predict Enrollment' button. The inputs are: 'Course Category' (dropdown menu), 'Course Price (\$)' (text input), 'Course Type' (dropdown menu), 'Duration (Hours)' (text input), 'Course Level' (dropdown menu), 'Course Rating' (slider), 'Instructor Expertise' (dropdown menu), and 'Instructor Experience (Years)' (text input). The 'Course Rating' slider is currently set to 4.00. The 'Predict Enrollment' button is located at the bottom of the form.

Field	Value
Course Category	Artificial Intelligence
Course Price (\$)	100.00
Course Type	Free
Duration (Hours)	20.00
Course Level	Advanced
Course Rating	4.00
Instructor Expertise	Artificial Intelligence
Instructor Experience (Years)	5

Figure 2. Enrollment Demand Prediction Module. Interactive form allowing users to specify course characteristics (category, price, level, rating, instructor expertise, duration) and receive estimated enrollment forecasts from the trained Random Forest model.

Navigation

Revenue Prediction

Revenue Forecasting

Course Category: Artificial Intelligence

Course Price (\$): 100.00

Course Type: Free

Duration (Hours): 20.00

Course Level: Advanced

Course Rating: 4.00

Instructor Expertise: Artificial Intelligence

Instructor Experience (Years): 5

Predict Revenue

Figure 3. Revenue Forecasting Module. Users can adjust course parameters interactively and generate revenue predictions using the optimized Ridge Regression model, enabling stress-testing of monetization strategies before live deployment.

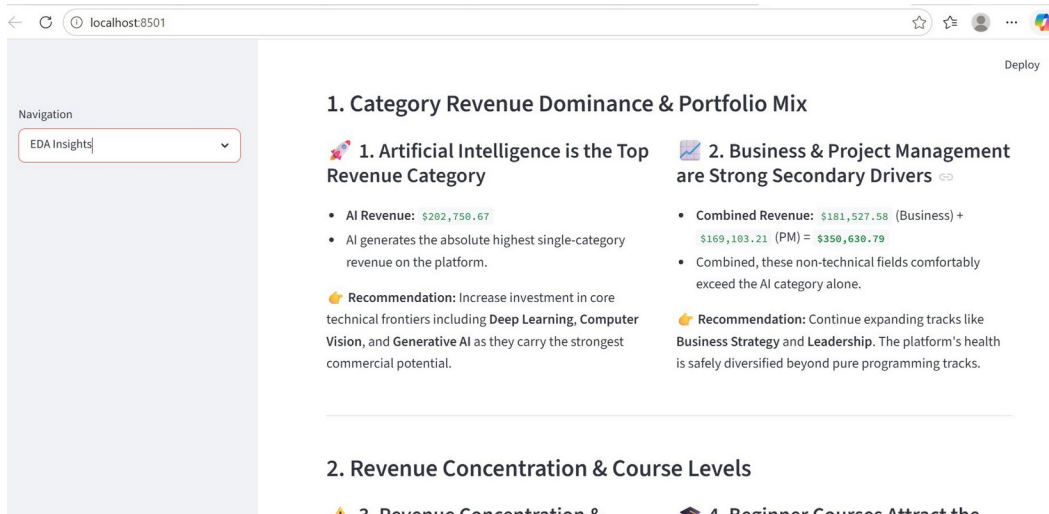


Figure 4. EDA Insights Module. Visualizes key exploratory findings including category revenue dominance (AI leading at \$202,750.67), Business & Project Management combined strength (\$350,630.79), and strategic portfolio recommendations.

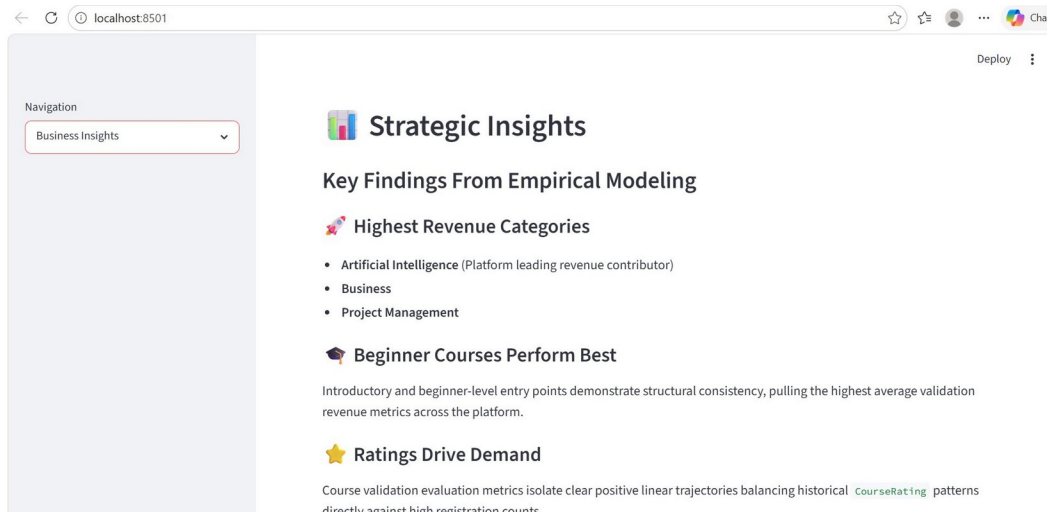


Figure 5. Strategic Insights & Business Recommendations. Synthesized findings from empirical modeling including highest-revenue categories (AI, Business, Project Management), beginner course performance metrics, and actionable recommendations for portfolio optimization and pricing strategy.

The Streamlit dashboard converts the research findings into an operational decision-support system. Rather than relying solely on static reports, decision-makers can explore scenarios interactively and immediately observe the predicted impact on enrollment demand and revenue outcomes. This bridges the gap between machine learning research and practical business implementation.

8.4 Potential Organizational Impact

If deployed within a production environment, the proposed framework could support multiple operational decisions. Enrollment forecasts may assist with instructor allocation, scheduling, and capacity planning. Revenue forecasts can support pricing decisions and investment prioritization. Category-level analysis can help management identify underperforming segments and allocate resources toward high-growth areas. Collectively, these capabilities allow the organization to transition from reactive reporting toward proactive data-driven decision-making.

9. Limitations and Future Work

Several limitations should be considered when interpreting the results of this study. First, the final modeling dataset consisted of 60 aggregated course records. Second, enrollment counts exhibited relatively low variability across courses, with a mean of 166.67 enrollments and a standard deviation of only 12.52, which limited predictive signal. Third, the modeling framework relied on a randomized 80:20 train-test split after transactions were aggregated to the course level, which removed the temporal sequence. Future work should retain transaction-level temporal information and implement time-based validation procedures to reduce look-ahead

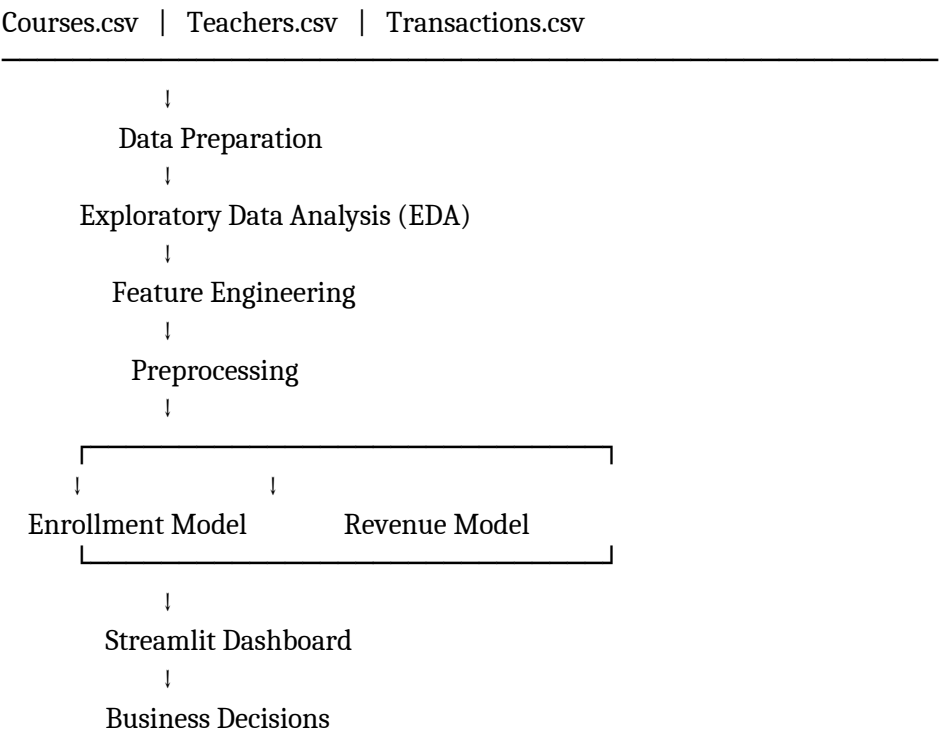
bias. Finally, incorporating learner demographics, marketing campaigns, geographic factors, or engagement metrics may significantly improve future forecasting performance.

Future versions of the system could integrate advanced machine learning approaches such as XGBoost, LightGBM, deep neural networks, and automated machine learning (AutoML) frameworks. Additionally, real-time recommendation systems and personalized learner analytics could be incorporated to further improve forecasting accuracy and business decision support.

9.1 Ethical Considerations

This study utilizes aggregated educational transaction data and does not rely on personally identifiable learner information. Future implementations should ensure compliance with applicable privacy regulations and institutional data governance policies. Any deployment involving learner-level prediction should incorporate transparency, fairness assessment, and appropriate safeguards to prevent unintended bias.

Framework Architecture Diagram



- The Streamlit dashboard transformed analytical findings into a practical decision-support system.
- Artificial Intelligence generated the highest category revenue.
- Instructor experience had minimal influence.
- Course ratings were the strongest enrollment-related factor.
- Course price emerged as the strongest revenue driver.
- Enrollment forecasting was constrained by low variance in enrollment counts.

- Revenue forecasting achieved excellent performance ($R^2 = 0.985$).

10. Conclusion

This study demonstrates the value of predictive analytics for educational business decisions. By combining enrollment forecasting, course-level revenue prediction, and a separate category-level analysis within a leakage-controlled pipeline, EduPro can move from reactive reporting toward proactive planning. Course-level revenue forecasting was highly accurate (Ridge Regression, $R^2 = 0.985$), and the category-level analysis showed that average category price explains almost all variation in category revenue ($r = 0.9984$). The exceptional revenue accuracy stems from the structure of the dataset rather than the complexity of the algorithm, making pricing strategy the single most effective lever for increasing future revenue. Enrollment prediction was modest (Random Forest, $R^2 = 0.205$), a direct consequence of low variance in the outcome. The analysis yielded clear, defensible business insights regarding pricing, category performance, and the limited role of instructor experience, providing solid directions for future software integrations.

More broadly, the study demonstrates that successful predictive analytics depends not only on algorithm selection but also on data quality, feature engineering, leakage prevention, and careful interpretation of results. These findings provide a practical foundation for future educational analytics systems and data-driven decision-support tools.

References

- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, 29(5), 1189–1232.
- Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly Media.
- Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques* (3rd ed.). Morgan Kaufmann.
- Kuhn, M., & Johnson, K. (2019). *Feature Engineering and Selection: A Practical Approach for Predictive Models*. CRC Press.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Romero, C., & Ventura, S. (2010). Educational Data Mining: A Review of the State of the Art. *IEEE Transactions on Systems, Man, and Cybernetics*, 40(6), 601–618.

The pandas development team. pandas: Python Data Analysis Library. Documentation.

Witten, I. H., Frank, E., Hall, M. A., & Pal, C. J. (2016). *Data Mining: Practical Machine Learning Tools and Techniques* (4th ed.). Morgan Kaufmann.